HBC-2 Monitor
Instruction Manual



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Color palette

The monitor uses a 4-bit color palette, therefore 16 colors are available.

Available colors

Number	Hex	Color	RGB
0	0x0	Black	#000000
1	0x1	Blue	#0000AA
2	0x2	Green	#00AA00
3	0x3	Cyan	#00AAAA
4	0x4	Red	#AA0000
5	0x5	Magenta	#AA00AA
6	0x6	Brown	#AA5500
7	0x7	Light gray	#AAAAAA
8	0x8	Bright black	#555555
9	0x9	Bright blue	#5555FF
10	0xA	Bright green	#55FF55
11	0xB	Bright cyan	#55FFFF
12	0xC	Bright red	#FF5555
13	0xD	Bright magenta	#FF55FF
14	0xE	Bright yellow	#FFFF55
15	0xF	White	#FFFFFF

Video modes

The monitor has 2 video modes, to enable pixel-perfect drawings or fast character display.

Available video modes

ID	Mode	Description
0	PIXEL	256 * 192 px 4-bit color mode
1	TEXT	6 * 8 px characters in a 42 columns by 24 rows grid Each with a 4-bit color **background** and a 4-bit color **font**

Colored text mode

This mode displays a **42 columns by 24 rows** grid of characters. **95 characters** can be displayed, corresponding to ASCII codes **32 to 127** (included, see the ASCII table below).

Each character has a **4-bit color background**, and another **4-bit color font**.

Pixel mode

Displays the whole 256 * 192 pixels monitor with **4-bit colors** (see the palette above). In this mode, each byte of video memory represented 2 pixels.

This would use **24,576 bytes of memory**, but for simplification this code stores each pixel in a whole byte (in the HBC-2 IDE).

ASCII Table

In colored test mode, ASCII codes **32 to 127** are the only one recognised by the monitor, corresponding to:

```
!"#$%&'()*+,-./0123456789:;⟸⟹?
@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_'abcdefghijklmnopqrstuvwxyz{|}~
```

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Control

Like every peripheral, the monitor uses sockets connecting it to the IOD ports to communicate with the CPU.

List of available ports

ID	Port	Description
0	DATA_0	First data byte, used for color information in both video modes
1	DATA_1	Second data byte, used for **colored text mode only**
2	POS_X	X-axis position of the designated pixel or character
3	POS_Y	Y-axis position of the designated pixel or character
4	CMD	Command sent by the CPU

List of available commands

ID	Command	Description
0	NOP	No operation
1	WRITE	Changes pixel or character information on DATA_0 and DATA_1 ports, depending on POS_X and POS_Y
2	READ	Sends pixel or character information on DATA_0 and DATA_1 ports, depending on POS_X and POS_Y
3	SWITCH_TO_PIXEL_MODE	Switches to **pixel mode** ***WARNING :*** *Does not flush the character buffer*
4	SWITCH_TO_TEXT_MODE	Switches to **colored text mode** ***WARNING :*** *Does not flush the pixel buffer*

Any invalid command will result in **NOP**.